$$\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{1}{1\cdot 3} + \frac{1}{2\cdot 4} + \frac{1}{3\cdot 5} + \dots$$ 5 6 3

b) Use D'Alembert's Ratio Test to find the interval of convergence of the following series 5 6 3

$$1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

c) Test the convergence of the following series using Cauchy's Root Test: 6 4 4

$$\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$$

Q5. a) Prove that the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is conditionally convergent. 5 6 3

b) Determine the interval of convergence for the following power series . 5 6 3

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2}$$

c) Obtain the Maclaurin series for the function $f(x) = \sin(x)$. 6 6 3

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DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

USN No: ENG25CS0453

B. Tech II Semester End Examinations – May 2026

Course Title: C Programming for Problem Solving
Course Code: 25EN1114
Duration: 02 Hours 30 Minutes
Date: 25-05-2026
Time: 01:00 PM to 03:30 PM
Max Marks: 80

Note:
1. **Answer all Five Full Questions**
2. **Each Full Question carries 16 Marks**
3. **Draw neat sketches wherever necessary**
4. **Missing Data may be suitably assumed**

Q. No.	Question		Marks	BTL	COs
1.a.	**Design** an algorithm and flowchart to find biggest among three numbers.		05	L2	CO1

1.b.	Predict the output for the following code:
	#include <stdio.h>
	int main() {
	int a = 5, b = 10, c;
	c = a++ * --b + b / a;
	printf("%d %d %d", a, b, c);
	return 0;
	}
	3, 9, 46
	#include <stdio.h>
	int main() {
	int a = 5, b = 9, c = -8, d = 1,z;
	z=a && b > 10 && c || d
	printf("%d",z);
	return 0;
	}
	1
			06	L2	CO1

1.c.	**Write** a C program to read the name of the user, number of units consumed and print out the charges. An electricity board charges the following rates for the use of electricity: for the first 100 units 80 paise per unit: for the next 100 units 90 paise per unit: beyond 300 units Rs 1 per unit. If the total amount is more than Rs 400, then an additional surcharge of 15% of total amount is charged.		05	L3	CO1
2.a.	**Demonstrate** selection sort on an array of 5 integers [64, 25, 12, 22, 11] with intermediate steps.		05	L3	CO2

(P.T.O.)